

Health Insurance Advisor Skill

Mathematical Model Specification

Readable PDF reference for feature transformation, factor construction, weighting, scoring, filtering, uncertainty, and feedback-driven reranking.

1. Purpose of the Mathematical Model

The model is a decision-support framework for ranking health insurance plans based on user-specific needs, preferences, risks, and constraints. It is not a claim-approval predictor, not a medical diagnosis tool, and not a legal conclusion engine.

The model combines three inputs: user information, derived risk and preference factors, and normalized plan data. It outputs ranked plan options, confidence labels, and transparent reasoning.

2. Core Notation

Let $X = \{x_1, x_2, \dots, x_n\}$ denote the raw user feature vector collected during intake.

Let $F = \{f_1, f_2, \dots, f_m\}$ denote the derived factor vector constructed from X .

Let $P = \{p_1, p_2, \dots, p_j\}$ denote the set of candidate health insurance plans.

Let $Z(p)$ denote the normalized plan-feature vector for plan p .

Let $W = \{w_1, w_2, \dots, w_k\}$ denote the weight vector used in ranking.

Let C denote the hard-constraint set. Let $S(p)$ denote the total score of plan p .

3. User Feature Space

The raw user vector X should include demographic, household, geographic, occupational, financial, health, medication, provider-preference, and utilization features.

Examples include age, ZIP, household size, employment type, monthly premium budget, chronic-condition presence, medication dependence, specialist frequency, doctor continuity preference, and risk tolerance.

Not every user must supply every variable. Missingness is permitted and carried forward into confidence scoring.

4. Derived-Factor Construction

The model maps raw features X into derived factors F through a transformation function $F = g(X)$. The purpose of $g(\cdot)$ is not to predict disease, but to compress user information into decision-relevant dimensions for plan comparison.

Example derived factors include care intensity, medication dependency, network dependency, financial fragility, catastrophic-cost sensitivity, mental-health-service dependency, occupational health stress, environmental access risk, and preference for cost predictability.

5. Example Derived-Factor Definitions

5.1 Care intensity

A simple ordinal model can be used: f_{care} = low, moderate, or high based on chronic-condition count, expected PCP frequency, specialist frequency, recent hospitalization, planned procedures, and expected lab/imaging intensity.

5.2 Medication dependency

f_{med} increases with the number of recurring medications, specialty-drug dependence, brand-only dependence, refill sensitivity, and the importance of formulary continuity.

5.3 Network dependency

f_{network} increases when the user must retain specific doctors, specialist systems, or hospital systems, or when the user has low willingness to switch providers.

5.4 Financial fragility

$f_{\text{fragility}}$ increases when premium budget is tight, deductible shocks are hard to absorb, emergency savings are limited, or the user prioritizes cost stability.

5.5 Catastrophic-cost sensitivity

f_{cat} increases when the user has low tolerance for worst-case annual exposure, elevated utilization uncertainty, or major downside-risk aversion.

6. Plan-Normalization Model

Every candidate plan p must be transformed into a normalized feature vector $Z(p)$. A normalized plan vector may include premium, deductible, out-of-pocket maximum, network type, provider-compatibility flags, formulary fit, mental-health access, referral friction, HSA eligibility, telehealth access, and evidence availability.

A generic plan vector can be written as:

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Z(p) = [premium(p), deductible(p), oopmax(p), networkfit(p), drugfit(p), mhfit(p), flexibility(p), adminsimplicity(p), evidencequality(p)]
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7. Constraint Model

Hard constraints are conditions that disqualify or strongly penalize plans. Examples include: plan not offered in the user's geography, plan cost above the user's absolute ceiling, must-have medication not covered, or must-have doctor not in network.

Define an eligibility indicator $I(p)$:

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I(p) = 1 if plan p satisfies all hard constraints in C; otherwise I(p) = 0
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If $I(p) = 0$, either exclude p from ranking or impose a severe penalty large enough to push the plan below viable options.

8. Weight Construction Model

The ranking model begins with a base weight vector W_0 . This vector is adjusted by derived factors and explicit preferences.

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W = Normalize(W0 + A(F) + B(Pref) + H(Constraints))
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where $A(F)$ represents factor-driven adjustments, $B(Pref)$ represents user-stated preference overrides, and $H(Constraints)$ encodes any scoring-level constraint adjustments.

A simple base vector may allocate weight across affordability, expected annual cost, catastrophic protection, provider fit, drug fit, mental-health fit, flexibility, and administrative simplicity.

9. Example Weight Dimensions

Let the ranking dimensions be indexed by k in $\{1, \dots, K\}$. One practical structure is:

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W = [wA, wE, wC, wN, wD, wM, wF, wS, wP]
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where w_A = affordability, w_E = expected annual cost, w_C = catastrophic protection, w_N = network/provider fit, w_D = drug/formulary fit, w_M = mental-health fit, w_F = flexibility, w_S = administrative simplicity, and w_P = preference alignment.

10. Component-Scoring Model

For each plan p , compute a component score vector $R(p)$:

$$R(p) = [r_A(p), r_E(p), r_C(p), r_N(p), r_D(p), r_M(p), r_F(p), r_S(p), r_P(p)]$$

Each component score should be normalized to a common scale, for example $[0,1]$ or $[0,100]$, so that weighted aggregation is meaningful.

10.1 Affordability score

A simple affordability score can be computed from premium relative to budget. For example, if premium is comfortably below budget, affordability score rises; if premium exceeds the soft budget, affordability score falls. The function should be monotone decreasing in premium burden.

10.2 Expected annual cost score

This score reflects annual premium plus expected out-of-pocket costs under the user's anticipated utilization pattern. Lower expected total annual cost implies a higher score.

10.3 Catastrophic protection score

This score reflects downside protection in bad years, typically increasing when the deductible and out-of-pocket maximum are lower relative to the user's financial fragility and catastrophic-cost sensitivity.

10.4 Provider-fit score

This score increases if must-keep providers are in network, specialist access is adequate, and network breadth aligns with user preferences.

10.5 Drug-fit score

This score rises when recurring medications are covered on favorable tiers and when prior-authorization or specialty restrictions are less burdensome.

10.6 Mental-health-fit score

This score increases when therapy, psychiatry, telehealth mental-health access, and network adequacy align with user needs.

10.7 Flexibility score

This score captures portability, out-of-area usability, PPO flexibility, lower referral friction, and the ability to accommodate changing provider needs.

10.8 Administrative-simplicity score

This score captures ease of use, lower administrative friction, and reduced likelihood of practical hassle, when such evidence exists.

10.9 Preference-alignment score

This score directly measures fit against the user's stated priorities, such as minimizing premium, preserving a doctor, or avoiding high yearly downside.

11. Total Plan Score

The total plan score can be written as the weighted sum of component scores:

$$S(p) = I(p) \times \sum_{k=1..K} w_k r_k(p)$$

If $I(p) = 0$, the plan fails the hard-constraint screen. If $I(p) = 1$, the plan remains viable and is ranked by $S(p)$.

12. Ranking Rule

Let $P^* = \{p \text{ in } P : I(p) = 1\}$ be the viable plan set. The final ranking is obtained by sorting viable plans in descending order of $S(p)$. The top 3 to 5 plans are then returned to the user.

$$\text{Rank} = \text{SortDescending}(\{S(p) : p \text{ in } P^*\})$$

13. Tie-Breaking and Near-Tie Handling

If two plans have very close scores, the system should not pretend the difference is decisive. Instead, it should label the result as a near tie and explain the qualitative difference between the alternatives, such as lower premium versus stronger protection.

If $|S(p_i) - S(p_j)| < \text{epsilon}$, label as near-tie and compare on qualitative tradeoffs

14. Expected-Cost Approximation

A simplified expected annual cost model may be used when no detailed claims simulation exists:

$EAC(p) = \text{AnnualPremium}(p) + \text{ExpectedOOP}(p \mid \text{user profile})$

ExpectedOOP can be approximated from visit frequency, specialist frequency, likely labs, medication burden, and the plan's cost-sharing structure. This estimate should be treated as approximate, not exact.

15. Downside-Risk Approximation

A simple downside-risk metric may be defined as a function of deductible, out-of-pocket maximum, and the user's fragility factor:

$DR(p) = \alpha \times \text{deductible}(p) + \beta \times \text{oopmax}(p)$ weighted by fragility and catastrophic sensitivity

The catastrophic-protection score is then inversely related to $DR(p)$. Lower downside risk implies stronger protection.

16. Preference-Override Model

When the user explicitly states a strong preference, this should modify the weight vector directly. For example, a user who says "keeping my doctor matters most" should receive a larger network-fit weight even if other inferred signals are weak.

$W' = \text{Normalize}(W + \Delta_{\text{pref}})$

17. Feedback-Driven Reranking

The framework itself remains fixed. User feedback updates only weights, constraints, or assumptions. The model then reruns the same ranking logic.

$W_{\text{new}} = \text{UpdateWeights}(W, \text{feedback})$
$C_{\text{new}} = \text{UpdateConstraints}(C, \text{feedback})$
$S_{\text{new}}(p) = I_{\text{new}}(p) \times \sum_{k=1..K} W_{k, \text{new}} r_k(p)$

This creates a principled reranking loop rather than an ad hoc change of reasoning framework.

18. Confidence Model

Recommendation confidence should depend on completeness of profile information, quality of plan data, and the degree of score separation among top plans.

$\text{Confidence} = h(\text{profile completeness, data quality, constraint certainty, score separation})$
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High confidence requires complete or near-complete user information, reliable evidence for major plan fields, and stable top-plan separation. Low confidence applies when key facts are missing or the top ranking is highly sensitive to assumptions.

19. Missing-Information Penalty

Missing critical information should lower confidence and may also reduce the stability of some component scores. If provider continuity is central but doctor names are unknown, provider-fit scoring must be treated as provisional.

$\text{ConfAdj} = \text{BaseConf} - \text{penalty}(\text{missing critical fields}) - \text{penalty}(\text{conflicting answers})$
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20. Governance Constraints on the Mathematics

The mathematical model must not create false precision or unlawful inference. Sensitive variables should be handled cautiously, and some should affect questioning rather than direct scoring. The model must separate documented facts, derived factors, and speculation.

Examples of prohibited misuse include claiming that a plan will definitely cover a claim, treating brochure language as definitive contract language, or using sensitive traits in discriminatory ways.

21. Minimal End-to-End Model Summary

The overall mathematical framework can be summarized in six steps:

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| 1. Collect user features X |
| 2. Derive factor vector $F = g(X)$ |
| 3. Build weights and constraints W, C |
| 4. Normalize each plan into $Z(p)$ |
| 5. Compute total score $S(p)$ |
| 6. Rank plans and update via feedback if needed |

22. Practical Interpretation

This model is best understood as a transparent multi-objective decision model. It is closer to a weighted risk-and-fit optimization framework than to a pure actuarial or predictive model. Its role is to discipline decision-making, force explicit tradeoffs, and support consistent reranking when user preferences change.